# 16S rDNA-Based Amplicon Analysis Unveiled a Correlation Between the Bacterial Diversity and Antibiotic Resistance Genes of Bacteriome of Commercial Smokeless Tobacco Products

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ORIGINAL ARTICLE

# 16S rDNA-Based Amplicon Analysis Unveiled a Correlation Between the Bacterial Diversity and Antibiotic Resistance Genes of Bacteriome of Commercial Smokeless Tobacco Products

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**Abstract**

The distribution of bacterial-derived antibiotic resistance genes (ARGs) in smokeless tobacco products is less explored and encourages understanding of the ARG profile of Indian smokeless tobacco products. Therefore, in the present investigation, ten commercial smokeless tobacco products were assessed for their bacterial diversity to understand the correlation between the inhabitant bacteria and predicted ARGs using a 16S rDNA-based metagenome analysis. Overall analysis showed the dominance of two phyla, i.e., Firmicutes (43.07%) and Proteobacteria (8.13%) among the samples, where *Bacillus* (9.76%), *Terribacillus* (8.06%), *Lysinibacillus* (5.8%), *Alkalibacterium* (5.6%), *Oceanobacillus* (3.52%), and *Dickeya* (3.1%) like genera were prevalent among these phyla. The phylogenetic investigation of communities by reconstruction of unobserved states (PICRUSt)-based analysis revealed 217 ARGs which were categorized into nine groups. Cationic antimicrobial polypeptides (CAMP, 33.8%), vancomycin (23.4%), penicillin-binding protein (13.8%), multidrug resistance MDR (10%), and β-lactam (9.3%) were among the top five contributors to ARGs. *Staphylococcus*, *Dickeya*, *Bacillus*, *Aerococcus*, and *Alkalibacterium* showed their strong and significant correlation ($p$ value < 0.05) with various antibiotic resistance mechanisms. ARGs of different classes (*blaTEM*, *blaSHV*, *blaCTX*, *tetX*, *vanA*, *aac3-II*, *mcr-1*, *intI-1*, and *intI2*) were also successfully amplified in the metagenomes of SMT samples using their specific primers. The prevalence of ARGs in inhabitant bacteria of smokeless tobacco products suggests making steady policies to regulate the hygiene of commercial smokeless tobacco products.

**Keywords** Smokeless tobacco · Antibiotic resistance genes · Microbiome · Policy · Next-generation sequencing

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## Introduction

Smokeless tobacco (SMT) is a non-combustible tobacco product that constitutes ~ 4000 chemical compounds [1]. Of them, 30 chemicals belong to the category of carcinogens [2]. More than 300 million consumers of smokeless tobacco products exist worldwide with the major chunk of > 85% only from South-Asian countries [3]. India's share of tobacco users is significantly high, where 20 million adults including men and women consume some form of smokeless tobacco regularly [4]. Daily intake of tobacco elevates the risk of oral cancer and other oral diseases in humans [5]. Besides the toxicants and carcinogens, inhabitant microorganisms of smokeless tobacco products have also been reported to pose risks to human health [6, 7]. The associated microbes of smokeless tobacco products exhibit the potential to alter the oral microbiome and cause several oral diseases [8]. Srivastava et al. (2021) reported an elevated count of *Staphylococcus*, *Fusobacterium*, and *Campylobacter* in the oral cavity of oral squamous cell carcinoma (OSCC) patients having smokeless tobacco chewing habits [9]. In our previous investigation on smokeless tobacco microbiome from Indian samples, we reported that smokeless tobacco products harbor bacteria engaged in nitrate/nitrite reduction, biofilm formation, and pro-inflammation activity [10]. Besides, a strong correlation was also observed between these activities and pathogenic bacteria. It indicates that commercial smokeless tobacco products pose a high bacterial-derived risk to human health. Sajid et al. also reported that smokeless tobacco bacteriome exhibits genes involved in toxin production, pro-inflammation activity, and nitrogen metabolism. Besides, genes involved in antibiotic drug resistance were also reported for the first time from the Indian smokeless tobacco products [11]. The occurrence of antibiotic-resistant bacteria in smokeless tobacco products further alarms its consumption to avoid the dissemination of ARG-resistant genes among oral or gut bacteriome of humans. There are only two reports that discuss the ARG profile from Indian smokeless tobacco products [11] which discuss only the relative abundance of the ARGs from indigenous tobacco products. Therefore, in the present investigation, we have attempted to explore the bacterial diversity of commercial smokeless tobacco products to understand the correlation between the inhabitant bacteria and antibiotic resistance genes (ARGs) using 16S rDNA-based metagenome analysis.

## Methodology

### Sample Collection and Metagenomic DNA Extraction

Ten popular smokeless tobacco (SMT) products were collected from the local markets of various parts of North India and kept at 4 °C till use (Table 1). For metagenomic DNA isolation, 1.0 g of smokeless tobacco samples was suspended in 10-ml sterile phosphate buffer saline (0.1 M, pH 8.0) and kept for vertexing to lose the adhered bacterial cells followed by sonication for 5 min (30 s on/off) in an ice bucket to harbor the endophytic bacterial communities from the leaves. The samples were centrifuged at a high speed of 10,000 rpm to collect the bacterial pellet present in the sample. The pellet was treated with 2.5 µl of lysozyme (10 mg/ml), proteinase-K (10 mg/ml), and 4.0-ml extraction buffer [*N*-cetyl,*N*,*N*,*N*-trimethylammonium bromide 1%, polyvinylpolypyrrolidone 2%, 1.5 M NaCl, 100 mM EDTA, 0.1 M TE buffer (pH 8.0), 0.1 M sodium phosphate buffer (pH 8.0), and 100 µl RNase A solution (10 mg/ml)] as

**Table 1** Detail of smokeless tobacco product information, pH, and moisture content of all SMT samples

| S. no | Sample code | Type of tobacco | Place of collection | Collection time | Latitude and longitude | Leaf appearance (phenotype) | pH | Moisture (%) |
|---|---|---|---|---|---|---|---|---|
| 1 | SMT-1 | CT | Delhi, India | August, 2021 | 28.7041 N, 77.1025 E | Brown | 5.5 | 10.63 |
| 2 | SMT-2 | CT | Delhi, India | August, 2021 | 28.7041 N, 77.1025 E | Greenish yellow | 5.8 | 6.49 |
| 3 | SMT-3 | CT | Bihar, India | August, 2021 | 25.9644 N, 85.2722 E | Greenish brown | 6.1 | 6.87 |
| 4 | SMT-4 | CT | Delhi, India | August, 2021 | 28.7041 N, 77.1025 E | Greenish yellow | 5.7 | 16.32 |
| 5 | SMT-5 | CT | Lucknow, India | August, 2021 | 26.8467 N, 80.9462 E | Greenish yellow | 6.0 | 7.05 |
| 6 | SMT-6 | CT | Lucknow, India | August, 2021 | 26.8467 N, 80.9462 E | Brown | 6.3 | 8.43 |
| 7 | SMT-7 | CT | Lucknow, India | August, 2021 | 26.8467 N, 80.9462 E | Greenish yellow | 6.7 | 11.6 |
| 8 | SMT-8 | CT | Lucknow, India | August, 2021 | 26.8467 N, 80.9462 E | Brown | 8.7 | 21.1 |
| 9 | SMT-9 | CT | Kanpur, India | August, 2021 | 26.4499 N, 80.3319 E | Greenish brown | 6.4 | 9.97 |
| 10 | SMT-10 | CT | Lucknow, India | August, 2021 | 26.8467 N, 80.9462 E | Brown | 5.5 | 7.53 |

*CT* commercial tobacco, *Lat* latitude, *Long* longitude described by Verma and Satyanarayana [12]. The suspension was vortexed and incubated at 37 °C for 1 h followed by the addition of 100 µl of 10% SDS (w/v) and kept at 60 °C for another 1 h to lyse the bacterial cells. The lysate was cooled to room temperature and processed for purification from proteins by adding an equal volume of phenol::chloroform::isoamyl (25:24:1). The aqueous layer was treated with 0.7 V isopropanol for 1 h to precipitate the metagenomic DNA. The metagenomic DNA was collected by centrifugation at room temperature for 10 min. The DNA pellet was washed with 70% (v/v) ethanol and centrifuged for 5 min at 10,000 rpm followed by drying at room temperature. Finally, the dried DNA pellet was dissolved in 50 µl of sterile 0.1 M TE buffer (pH 8.0) and kept at − 20 °C until their use. The DNA concentration and purification ($A_{260}/A_{230}$ and $A_{260}/A_{280}$) assessed the quality of extracted metagenomes from respective samples (Supplementary Table 1).

## 16S Amplicon Library Preparation and Processing of Raw Reads

Five nanograms of qubit quantified metagenomic DNA from each sample was used for preparing the 16S amplicon library. The NEBNext ultra-DNA library preparation kit was used for the generation of the amplicon using bacterial-specific $V_3$–$V_4$ hypervariable regions ($V_3$F_*CCTACGGGNBGCASCAG* and $V_4$R_*GACTACNVGGGTATCTAATCC*). The equimolar concentration of each library was sequenced at the Illumina HiSeq2500 platform using the outsourcing facility of AgriGenome Labs Pvt. Ltd., Cochi, India. The paired-end raw reads were analyzed for various parameters such as QC, average base content per read, and G + C percentage (Supplementary Table 2). The raw reads were imported on the QIIME2 pipeline (ver. 2020.11). The reads of Phred score value > 30 were achieved through various plugins such as denoiser, chimera removal, and *q2-dada2* to obtain the amplicon sequence variants (ASVs). These ASVs were used for taxonomy assignment by using the QIIME feature classifier command with the SILVA-138 16S database [13].

## Bacterial Diversity and Antibiotic Resistance Gene (ARG) Profiling

Various diversity indices were calculated using the QIIME2 tool. The phylogenetic investigation of communities by reconstruction of unobserved states (PICRUSt2, v 2.4.1) tool [14] was used for analyzing antibiotic resistance gene (ARG) profiling.

## PCR Screening for Antibiotic Resistance Genes and Class 1 and 2 Integrons

Metagenomes of SMT samples were screened for the presence of different classes of antibiotics for the detection of antibiotic resistance genes (ARGs). It includes the genes encoding tetracycline (*tet-X*), vancomycin (*van-A*), mobile colistin resistance (*mcr-1*), quaternary ammonium compound (*aac3-II*), and three β-lactam resistance genes (*blaCTX-M*, *blaSHV*, and *blaTEM*). Besides, integron 1 (*intl1* and *intl2*) was also investigated in the metagenomic DNA samples of SMT products.

The PCR was performed in a 50-µl PCR reaction mixture containing 40 ng of metagenomic DNA, 25 µl of 2X PCR master mix (GeneDireX, 230 OnePCR™ Plus), and 1 µl of 10 pmol of respective primers (Table 2). The reaction was set as initial denaturation at 95 °C followed by 30 cycles of denaturation (1 min at 95 °C), annealing (30 s at 50–60 °C), and extension (1 min at 72 °C). A final extension of 10 min was given at 72 °C. The amplicons were run at 12% (w/v) agarose gel electrophoresis and visualized under the UV transilluminator.

**Table 2** Primers and annealing temperatures of various ARGs used in this investigation

| Target ARGs | Forward primer sequences (5′–3′) | Reverse primer sequences (5′–3′) | Amplicon size (bp) | Annealing temperature (°C) | Reference |
|---|---|---|---|---|---|
| *blaTEM* | ATGAGTATTCAACATTTCCGC | CAATGCTTAATCAGTGAGG | 856 | 53 | [15] |
| *blaSHV* | AGCCGCTTGAGCAAATTAAAC | GTTGCCAGTGCTCGATCAGC | 786 | 55 | [16] |
| *blaCTX* | GTTACAATGTGTGAGAAGCAG | CCGTTTCCGCTATTACAAAC | 593 | 50 | [17] |
| *vanA* | GCTGCGATATTCAAAGCTCA | CAGTACAATGCGGCCGTTA | 545 | 50 | [18] |
| *Mcr1* | CAGTATGGGATTGCGCAATGATT | TTATCCATCACGCCTTTTGAGTC | 1197 | 68 | [19] |
| *tetX* | TTAGCCTTACCAATGGGTGT | CAAATCTGCTGTTTCATTCG | 468 | 55 | [20] |
| *Aac3-II* | ACTGTGATGGGATACGCGTC | CTCCGTCAGCGTTTCAGCTA | 237 | 60 | [21] |
| *intI1* | GCCGCCAATGCCTGACGAT | TATCGCTGTATGGCTTCA | 700 | 55 | [22] |
| *intI2* | CGGGATCCCGGACGGCATGC-ACGATTTGTA | GATGCCATCGCAAGTACGAG | 650 | 55 | [23] |

### Statistical and Network Analysis

The Principal Coordinate Analysis (PCoA) plot and weighted unifrac distances were generated through QIIME2 (ver. 4.03, https://folk.uio.no/ohammer/past) to determine the diversity level in bacterial communities among ten SMT samples. The rarefaction curves were generated with a sampling depth of 1000 from each sample (Supplementary Fig. 1). Further, the Spearman correlation ($r_s$) was assessed between the genera (> 0.1% abundance) and functional genes involved in ARG pathways by using *otu.association* command [*mothur > out.association (Shared = file name.shared, method = spearman)*] in MOTHUR [24]. The Cytoscape (ver. 3.9.1 [25], https://cytoscape.org/) was used to visualize the correlation network. OrginPro (ver. origin 2024 (10.1 SR1), https://www.originlab.com/origin) was used to generate the heat map and visualize the correlation network.

## Results

### Taxonomic Assignment

A total of 403,346,0 raw reads were generated through the Illumina platform. The DADA2 plugin reduced the ASVs to 207,054,8 quality reads followed by taxonomy assignment using QIIME2 (Supplementary Table 3). Overall, these ASVs were assigned to 28 phyla, 54 classes, 108 orders, 154 families, 175 genera, and 87 species. Besides, 67 uncultured species and 30 uncultivable genera were also observed during the overall analysis.

### Dominant and Shared Bacterial Diversity

**Phylum-Level Analysis** The combined data set of all respective sequences from each sample revealed the dominance of Firmicutes (43.07%) and Proteobacteria (8.13%) among the various samples. Actinobacteria (0.8%), Bacteroidetes (0.15%), Cyanobacteria (0.02%), Acidobacteriota (0.018%), Planctomycetota (0.017%), Gemmatimonadota (0.008%), Methylomirabilota (0.006%), and Myxococcota (0.005%) were observed as the small contributors of SMT microbiome (Supplementary file 1, Fig. 1A).

Four different phyla, i.e., Firmicutes, Proteobacteria, Actinobacteriota, and Bacteroidota, were shared among all the SMT samples. Cyanobacteria was observed among 90% of the samples, whereas the members of Acidobacteriota and Planctomycetota showed their presence among 80% of SMT samples (Supplementary file 4, Fig. 1D).

**Genus-Level Analysis** A total of 175 assigned genera were obtained. Of them, the top seven dominant genera were *Bacillus* (9.76%), *Terribacillus* (8.06%), *Lysinibacillus* (5.85%), *Alkalibacterium* (5.6%), *Oceanobacillus* (3.52%), and *Dickeya* (3.1%) which showed their abundance > 1%, whereas *Aerococcus* (0.72%), *Corynebacterium* (0.45%), *Staphylococcus* (0.33%), *Stenotrophomonas* (0.26%), *Paenibacillus* (0.25%), *Paucisalibacillus* (0.23%), *Halomonas* (0.21%), *Pseudomonas* (0.17%), *Gracilibacillus*

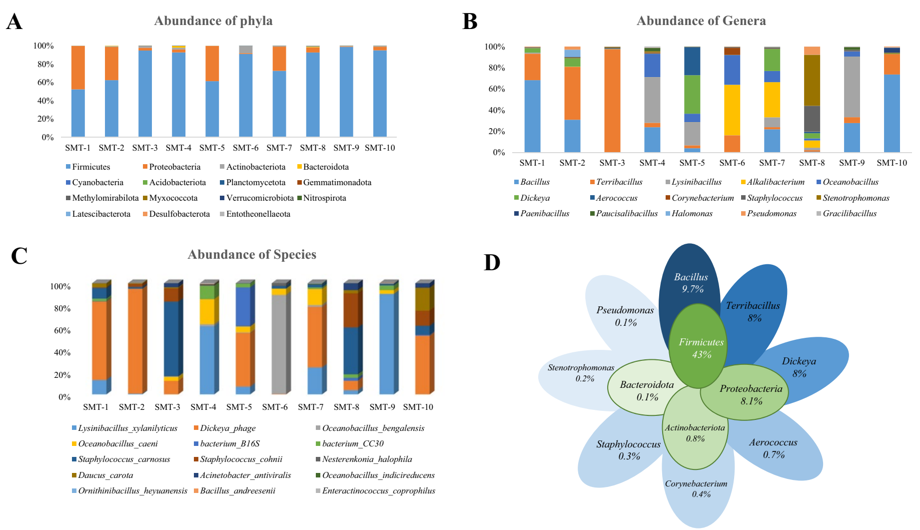


**Fig. 1** Relative abundance of bacteria (top 15) at phylum (**A**), genus (**B**), and species (**C**) level. Core bacterial microbiome of SMT products at phylum and genus level (**D**)

(0.11%), *Brachybacterium* (0.09%), *Glutamicibacter* (0.08%), *Facklamia* (0.06%), *Jeotgalicoccus* (0.05%), *Brevibacterium* (0.05%), and *Sphingobacterium* (0.05%) showed their abundance between 0.5 and 1% of the total genera obtained during the analysis (Supplementary file 2, Fig. 1B). Of the top 15 assigned genera (<0.1%), eight genera, i.e., *Bacillus, Terribacillus*, *Dickeya*, *Aerococcus*, *Corynebacterium*, *Staphylococcus*, *Stenotrophomonas*, and *Pseudomonas*, were common among all the samples. *Lysinibacillus*, *Alkalibacterium*, *Oceanobacillus*, and *Halomonas* were shared in more than 90% of SMT samples. *Paenibacillus* and *Gracilibacillus* were observed in more than 50% of the samples, whereas *Paucisalibacillus* was shared among 40% of the samples (Supplementary file 4, Fig. 1D).

**Species-Level Analysis** Of the 87 assigned species, only seven species showed their abundance > 0.1%. It includes *Lysinibacillus* xylanolytic (5.11%), *Dickeya phage* (3.1%), *Oceanobacillus bengalensis* (1.56%), *Oceanobacillus caeni* (1.07%), *Bacterium B16S* (0.71%), *Bacterium CC30* (0.41%), and *Staphylococcus carnosus* (0.33%), whereas *Staphylococcus cohnii* (0.09%), *Daucus carota* (0.02%), *Acinetobacter antiviralis* (0.01%), *Oceanobacillus indicireducens* (0.01%), *Ornithinibacillus heyuanensis* (0.01%), *Bacillus_andreesenii* (0.01%), *Enteractinococcus_coprophilus* (0.01%), and *Pseudomonas geniculate* (0.01%) were also observed that showed their abundance between 0.1 and 0.01% (Supplementary file 3, Fig. 1C). Of the top five species (< 0.5% abundance) *Dickeya_phage* and *bacterium_B16S* were present among all the samples, whereas *Lysinibacillus xylanilyticus* was observed among 80% of the SMT samples. *Oceanobacillus caeni* and *Oceanobacillus bengalensis* were shared in 60% and 40% of the SMT samples. *Staphylococcus carnosus* and *Staphylococcus cohnii* were observed among all the samples; however, their abundance was non-significant (*p* value > 0.1–0.05%) (Supplementary file 4).

## Inter-individual Differences

**Phylum-Level Analysis** Though the bacterial composition at the phylum level was highly conserved in this investigation, inter-individual differences were observed at the phylum, genus, and species level. Phylum-level comparative analysis of the top five phyla (Firmicutes, Proteobacteria, Actinobacteriota, Bacteroidota, and Cyanobacteria) unveiled that Firmicutes were dominant among all the samples. Sample SMT-8 exhibited the maximum count of Firmicutes. Proteobacteria was observed as the second major contributor among all the STPs except SMT-6, where Actinobacteriota was the second major contributor.

**Genus-Level Analysis** Inter-individual differences at the genus level resulted in a very similar pattern as obtained during the analysis at the phylum level. Two samples (SMT-1 and SMT-10) were populated by *Bacillus*, whereas *Terribacillus* showed its prevalence in SMT-2 and SMT-3 samples. Similarly, *Lysinibacillus* also showed their dominance in two samples (SMT-4 and SMT-9). *Bacillus* showed its dominance in the second order in four samples (SMT-2, SMT-4, SMT-7, and SMT-9). Similarly, another close genus of *Bacillus*, i.e., *Terribacillus*, showed second-order abundance in two samples (SMT-1 and SMT-10). SMT-6 and SMT-7 were dominated by *Alkalibacterium*; *Dickeya*, *Stenotrophomonas*, *Oceanobacillus*, and *Aerococcus* were observed as dominant genera in only one sample, i.e., SMT-5, SMT-8, SMT-6, and SMT-5, respectively. *Staphylococcu*s also showed second-order dominance only in two samples (SMT-3 and SMT-8).

**Species-Level Analysis** *Dickeya_phage* showed their dominance in four samples (SMT-2, SMT-5, SMT-7, and SMT-10), while three samples (SMT-1, SMT-4, and SMT-9) were populated by *Lysinibacillus_xylanilyticus*. *Staphylococcus_carnosus* showed its dominance in two samples only, i.e., SMT-3 and SMT-8, whereas SMT-6 was highly populated with *Oceanobacillus bengalensis*. This species showed second-order dominance in three samples (SMT-4, SMT-6, and SMT-7). Similarly, SMT-1 and SMT-3 were populated with *Dickeya_phage* at the second-order species, whereas *Staphylococcus_cohnii* also showed their second-order dominance in the other two samples, i.e., SMT-2 and SMT-8. *Bacterium_B16S*, *Bacterium_CC30*, and *Daucus_carota* were identified as the top dominant species in one sample, i.e., SMT-5, SMT-9, and SMT-10, respectively.

## Diversity Indices

Sample SMT-4 showed maximum richness by having maximum diversity indices [Shannon (7.03), evenness vector (0.73), observed features (798), and Simpson index (0.98)], while maximum Faith_pd was observed in SMT-1 (365.89); the remaining SMT samples showed a Faith_pd index range between 111.74 and 183.75 which represents the diversity relatedness among these samples. Maximum Chao1 value was observed in SMT-8 (947.88), while SMT-10 showed minimum Chao1 (450.25). SMT-3 represents the minimum Simpson (0.731783) and Shannon (3.3068) values. All the SMT samples showed > 99% Good's coverage, which indicates that 100 more sequences will add merely one ASV (Table 3). β-diversity through the Bray–Curtis dissimilarity index showed the diversity among the samples, whereas the weighted_unifrac showed comparatively less diversity among the SMT samples (Fig. 2A, B).

**Table 3** Various diversity indices and estimators of bacterial diversity from smokeless tobacco products

| Sample Id | Dominant phyla | Dominant genera | Dominant species | Faith_pd vector | Shannon | Evenness vector | Simpson index | Chao1 | Observed_features | Good's estimator |
|---|---|---|---|---|---|---|---|---|---|---|
| SMT-1 | Firmicutes (25.2%) | *Bacillus (27.6%)* | *Lysinibacillus_xylanilyticus (0.1%)* | 356.8864 | 5.136356 | 0.550071 | 0.937255 | 705.3953 | 647 | 0.999915 |
| SMT-2 | Firmicutes (32.6%) | *Terribacillus (18.6%)* | *Dickeya_phage (2.9%)* | 166.1464 | 5.331996 | 0.58623 | 0.942451 | 579.1333 | 547 | 0.999858 |
| SMT-3 | Firmicutes (47.3%) | *Terribacillus (45.9%)* | *Staphylococcus_carnosus (0.5%)* | 118.2754 | 3.3068 | 0.375873 | 0.731783 | 491.0161 | 445 | 0.999823 |
| SMT-4 | Firmicutes (46.3%) | *Lysinibacillus (18.6%)* | *Lysinibacillus_xylanilyticus (14.6%)* | 171.3023 | 7.030061 | 0.730481 | 0.984261 | 803.8916 | 789 | 0.999727 |
| SMT-5 | Firmicutes (27.6%) | *Dickey (14.2%)* | *Dickeya_phage (14.2%)* | 111.7379 | 5.918581 | 0.655788 | 0.957894 | 540.814 | 521 | 0.999908 |
| SMT-6 | Firmicutes (49.9%) | *Alkalibacterium (24.6%)* | *Oceanobacillus_bengalensis (13.5%)* | 183.7489 | 4.520331 | 0.495991 | 0.888332 | 648.5588 | 554 | 0.999802 |
| SMT-7 | Firmicutes (38.6%) | *Alkalibacterium (15.8%)* | *Dickeya_phage (9.8%)* | 131.3463 | 6.11879 | 0.640532 | 0.963791 | 882.0435 | 751 | 0.999944 |
| SMT-8 | Firmicutes (47%) | *Stenotrophomonas (1%)* | *Staphylococcus_carnosus (0.5%)* | 134.4535 | 2.954567 | 0.309106 | 0.757834 | 947.8824 | 754 | 0.999827 |
| SMT-9 | Firmicutes (51.7%) | *Lysinibacillus (30.9%)* | *Lysinibacillus_xylanilyticus (30.2%)* | 158.5832 | 6.233044 | 0.677891 | 0.967328 | 631.3469 | 586 | 0.999949 |
| SMT-10 | Firmicutes (57.3%) | *Bacillus (38.5%)* | *Dickeya_phage (0.2%)* | 156.8977 | 3.611747 | 0.416294 | 0.791672 | 450.25 | 409 | 0.999943 |

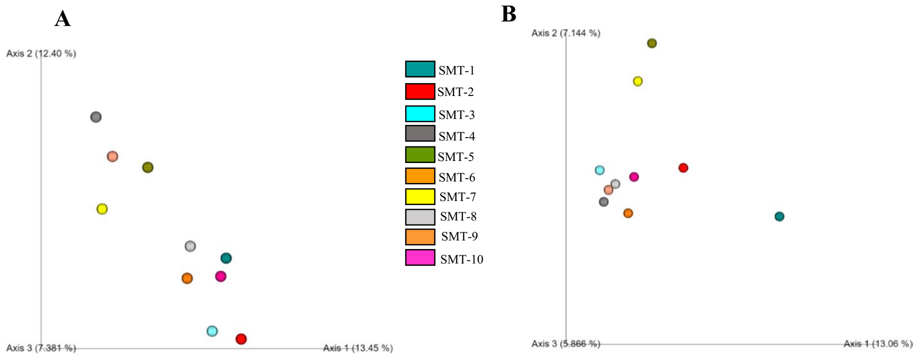


**Fig. 2** The PCoA plot showing the β-diversity through Bray–Curtis's dissimilarity indices (**A**) and weighted_unifrac dissimilarity index (**B**)

## Antibiotic Resistance Gene Profiling

A total of 217 ARGs were observed among 10 SMT samples which were categorized into nine different mechanisms. It includes CAMP (33.8%), vancomycin (23.4%), penicillin-binding protein (13.8%), multidrug resistance MDR (10%), β-lactam (9.3%), tetracycline (7.7), rifampicin (1%), quinolones (0.4%), and aminoglycoside/ase (0.1%) (Table 4, Fig. 3). Here, we have discussed only those ARGs that showed their relative abundance > 1% in their respective pathways to make the data more practical. It includes (i) CAMP (*amiABC*, *degP*, *dltA*, *dltD*, *dltC*, *dltB*, *arnE*, *acrA*, *mrF*, *dsbA*, *phoP*, *tolC*, *eptA*, *lpxA*, *pagP*, *PPIA*, *marA*, and *phoQ*), (ii) vancomycin (*alr*, *ddl*, *murG*, *mraY*, *murF*, *vanY*, and *vanX*), (iii) penicillin-binding protein (*mcrA*, *pbp1b*, *pbp2A*, *pbp5*, *mdrA*, *ftsl*, *mrcB*, and *pbpC*), (iv) MDR (*abcA*, *norG*, *acrB*, *acrA*, *tolC*, *ompF*, *marA*, *bpeT*, *oprM*, *norB*, *envR*, *acrE*, *acrF*, *evgA*, *adeR*, *adeS*, *adeB*, and *mexL*), (v) beta-lactam (*penP*, *blaOXA*, *blaSHV*, *blaTEM*, *blaCTX*, and *ampC*), (vi) tetracycline (*msrA*, *vga*, *ereA*, *steB*, *mph*, *catA*, *mdfA*, *mef*, *tetA*, *catB*, *sul1*, *tetB*, and *vgb*), (vii) rifampicin (*arr*), (viii) quinolones (*smvA* and *qnr*), and (ix) aminoglycoside (*aadK*, *aacC*, *aadA*, *aac6-I*, *aph3-II*, *aph3-I*, *acc3-II*, *aac6-1*, and *aph*) (Supplementary file 5).

## Correlational and Network Analysis

A correlation analysis was performed between the top 15 genera and different antibiotic resistance mechanisms obtained during this investigation (Supplementary file 6, Supplementary Fig. 2). Network analysis of top 15 genera and antibiotic resistance mechanism showed 24 nodes and 135 edges, and the network density was 0.245. Here, we have mentioned only those correlations ($r_s$) that were either significantly positive or negative (cutoff − 0.887 to + 0.887) between the studied genera and respective antibiotics. *Staphylococcus* showed a significant correlation with vancomycin (value $r_s$, 0.8), tetracycline (value $r_s$, 0.7), and rifampicin (value $r_s$, 0.6). *Dickeya* showed strong and significant correlation with quinolones (value $r_s$, 0.73), aminoglycoside/ase (value $r_s$, 0.61), and multidrug-resistant genes (value $r_s$, 0.61), whereas *Bacillus*, *Aerococcus*, and *Alkalibacterium* showed their strong and significant correlation with β-lactam (value $r_s$, 0.81), quinolones (value $r_s$, 0.69), and tetracycline (value

**Table 4** Relative abundance (%) of ARGs of SMT samples

| | Antibiotic resistance genes/samples | SMT products | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SMT-1 | SMT-2 | SMT-3 | SMT-4 | SMT-5 | SMT-6 | SMT-7 | SMT-8 | SMT-9 | SMT-10 |
| Different ARGs | β-lactam resistance | 6.04% | 8.4% | 12.6% | 4.2% | 13% | 8.3% | 28% | 1.5% | 9.5% | 8.1% |
| | Penicillin-binding protein | 6.1% | 7.5% | 8.6% | 4.1% | 6.3% | 12% | 23.7% | 15.1% | 10% | 5.9% |
| | CAMP | 8.3% | 8.8% | 10.8% | 3.9% | 10.6% | 6.6% | 27.6% | 5.2% | 9.2% | 8.5% |
| | Aminoglycoside | 10.1% | 8.1% | 19% | 2.9% | 16.9% | 7.8% | 25.5% | 1.1% | 4.5% | 3.6% |
| | Tetracycline resistance | 4.9% | 6.6% | 7.8% | 3.6% | 8.6% | 12.2% | 26.6% | 12.9% | 8.9% | 7.6% |
| | Quinolones | 5.5% | 3.1% | 2.1% | 1.2% | 31% | 3.8% | 46.5% | 4.3% | 0.8% | 1.1% |
| | Rifampicin | 11.8% | 0.8% | 0.3% | 1.5% | 0.5% | 19.5% | 20.1% | 43.8% | 1.1% | 0.1% |
| | Vancomycin | 6.8% | 6.1% | 7.7% | 5.1% | 5.3% | 12.3% | 20.9% | 15.4% | 13.2% | 6.7% |
| | MDR | 6.7% | 6.7% | 3.9% | 4% | 16% | 4.4% | 34.1% | 8.7% | 8.9% | 6.1% |

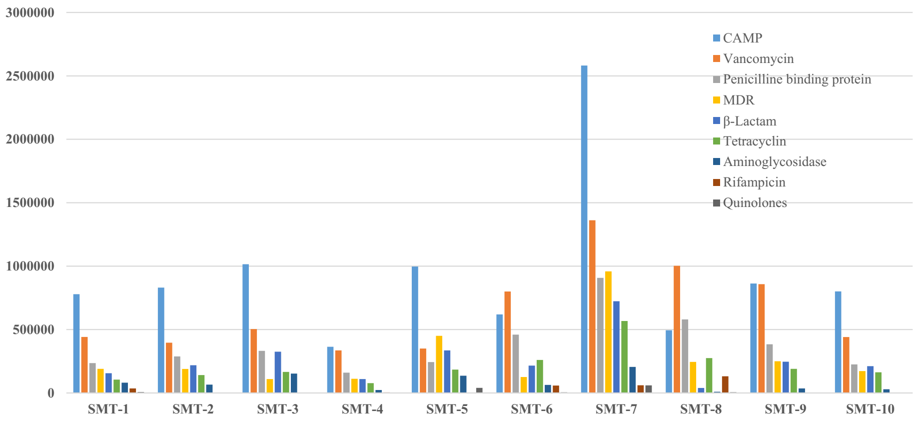


**Fig. 3** Relative abundance of different antibiotic resistance genes obtained through PICRUSt-based analysis among the inhabitant bacteria of different SMT products

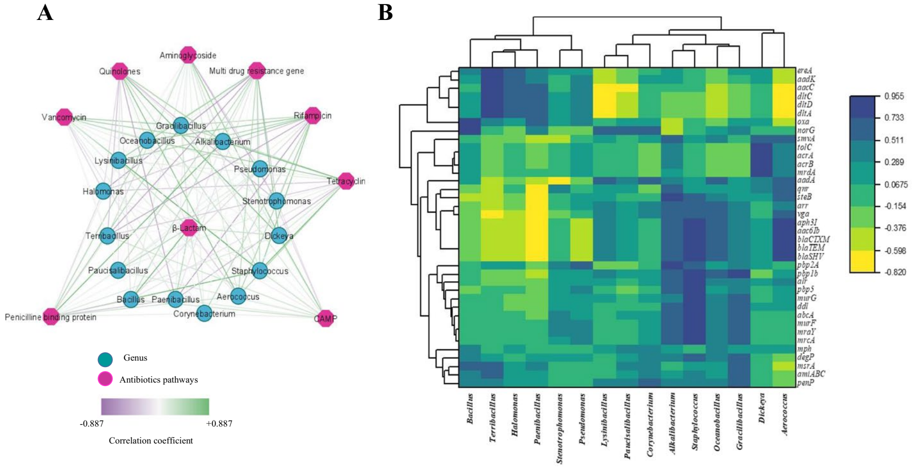


**Fig. 4** Spearman's correlation network analysis between the top 15 genera and nine different antibiotic resistance mechanisms (**A**). The heatmap showing significant correlation between the top 15 genera and top five ARGs (**B**)

$r_s$, 0.68), respectively. Genera *Halomonas* and *Paenibacillus* were negatively correlated with MDR-resistant genes (value $r_s$, −0.65) and rifampicin (value $r_s$, −0.61) (Fig. 4A).

Further, the top five genes of each ARG pathway were studied to correlate with the top 15 genera using heatmap correlation matrix analysis (Fig. 4B, Supplementary file 7). *Staphylococcus* showed a significant correlation with 15 different antibiotic-resistant genes [*murG* ($r_s$, 0.87), *murF* ($r_s$, 0.79), *pbp1b* ($r_s$, 0.83), *pbp5* ($r_s$, 0.79), *mraY* ($r_s$, 0.79), *mrcA* ($r_s$, 0.78), *alr* ($r_s$, 0.75), *blaSHV* ($r_s$, 0.74), *blaTEM* ($r_s$, 0.74), *blaCTX-M* ($r_s$, 0.74), *acc6Ib* ($r_s$, 0.74), *aph3I* ($r_s$, 0.74), *ddl* ($r_s$, 0.78), *abcA* ($r_s$, 0.76), and *arr* ($r_s$, 0.61)]. *Paenibacillus* showed strong and significant correlation with 13 different ARGs of which 4 ARGs [*dltA* ($r_s$, 0.70), *dltC* ($r_s$, 0.70), *dltD* ($r_s$, 0.70), and *oxa* ($r_s$, 0.61)] were significantly positive and 9 ARGs [*blaSHV* ($r_s$, −0.81), *blaTEM*

($r_s$, −0.81), *blaCTX-M* ($r_s$, −0.81), *aac6Ib* ($r_s$, −0.81), *aph3I* ($r_s$, −0.81), *vga* ($r_s$, −0.69), *steB* ($r_s$, −0.66), *qnr* ($r_s$, −0.61), and *arr* ($r_s$, −0.61)] were significantly negative, whereas *Aerococcus* showed a strong and significant correlation with 7 different ARGs [*blaSHV* ($r_s$, 0.85), *blaTEM* ($r_s$, 0.85), *blaCTX-M* ($r_s$, 0.85), *acc6Ib* ($r_s$, 0.85), *aph3I* ($r_s$, 0.85), *qnr* ($r_s$, 0.73), and *smvA* ($r_s$, 0.68)], while it was negatively correlated with *dltA* ($r_s$, −0.73), *dltD* ($r_s$, −0.73), *dltC* ($r_s$, −0.73), and *aacC* ($r_s$, −0.71). Genus *Alkalibacterium* showed a strong and significant correlation with 10 different ARGs [*steB* ($r_s$, 0.91), *pbp2A* ($r_s$, 0.75), *blaSHV* ($r_s$, 0.65), *blaTEM* ($r_s$, 0.65), *blaCTX-M* ($r_s$, 0.65), *aac6Ib* ($r_s$, 0.65), *aph3I* ($r_s$, 0.65), *pbp1b* ($r_s$, 0.64), and *mrcA* ($r_s$, 0.62)]. *Oceanobacillus* showed a positive and strong correlation with 7 ARGs [*aadA* ($r_s$, 0.70), *blaSHV* ($r_s$, 0.68), *blaTEM* ($r_s$, 0.98), *blaCTX-M* ($r_s$, 0.68), *aac6Ib* ($r_s$, 0.68), *aph3I* ($r_s$, 0.68), and *vga* ($r_s$, 0.63)]. *Terribacillus* showed a correlation with 6 different ARGs [*dltA* ($r_s$, 0.92), *dltD* ($r_s$, 0.92), *dltC* ($r_s$, 0.92), *aacC* ($r_s$, 0.91), *aadK* ($r_s$, 0.84), and *ereA* ($r_s$, 0.78)]. *Dickeya* exhibited a positive correlation with 5 different ARGs [*mrdA* ($r_s$, 0.95), *acrB* ($r_s$, 0.92), *acrA* ($r_s$, 0.89), *tolC* ($r_s$, 0.89), and *smvA* ($r_s$, 0.70)]. Genus *Halomonas* showed a strong correlation with four different ARGs (*dltA*, *dltD*, *dltC*, and *aadK*) with $r_s$ value 0.62. *Bacillus* showed a strong positive correlation with *norG* ($r_s$, 0.89) and *oxa* ($r_s$, 0.87) genes. *Pausibacillus* also showed a positive correlation with *norG* ($r_s$, 0.70), and it was negatively correlated with *aacC* ($r_s$, −0.61). *Corynebacterium* and *Stenotrophomonas* were strongly correlated with the *aadA* ($r_s$, 0.71 and −0.73) gene. *Gracilibacillus* showed significant correlation with *pbp1b* ($r_s$, 0.79), *penP* ($r_s$, 0.65), and *abcA* ($r_s$, 0.65) (Fig. 4B). Further genera *Lysinibacillus* and *Pseudomonas* showed negative and non-significant correlation with top 5% ARGs (Supplementary file 7).

### Screening of SMT Metagenomes for ARGs

Molecular confirmation of different ARGs among the metagenomic DNA of respective samples revealed that specific sequences of *blaTEM* were present among all the SMT samples, while *blaSHV* and *blaCTX-M* were observed in 90% (SMT-2, SMT-3, SMT-4, SMT-5, SMT-6, SMT-7, SMT-8, SMT-9, and SMT-10) and 30% (SMT-2, SMT-3, and SMT-10) of the SMT samples, respectively, whereas vancomycin and mobile colistin-resistant were also successfully amplified among all the SMT samples. *Tet-X* gene showed its presence in 70% (SMT-1, SMT-2, SMT-3, SMT-4, SMT-5, SMT-6, and SMT-9) of the total samples. ARG coding aminoglycoside/ase (*aac3-II*) was observed in 40% (SMT-5, SMT-6, SMT-7, and SMT-10) of SMT samples. The screening of mobile genetic elements integron class I (*intI1*) and class II (*intI2*) showed their presence among 70% (SMT-2, SMT-3, SMT-4, SMT-5, SMT-6, SMT-7, and SMT-8) and 60% (SMT-1, SMT-3, SMT-4, SMT-5, SMT-6, and SMT-9) samples only (Table 5, Fig. 5).

## Discussion

Regular consumption of SMT products has been associated with several perilous health illnesses [7]. Inhabitant bacteria of SMT products have also been correlated with altering healthy oral microbiome which may cause several oral diseases including oral cancer [8, 26]. In a recent investigation, several bacterial-derived health risks were studied from smokeless tobacco products [27]. This culture-dependent approach identified several bacteria from SMT products exhibiting biofilm formation, salt tolerance, and heavy metal tolerance activity. It indicates that tobacco-inhabitant bacteria also pose a significant risk to oral health. Besides, antibiotic resistance among bacteria is now a global threat

**Table 5** Amplification of various ARGs from the metagenomes of the SMT products in this investigation

| ARGs | SMT-1 | SMT-2 | SMT-3 | SMT-4 | SMT-5 | SMT-6 | SMT-7 | SMT-8 | SMT-9 | SMT-10 | Relative abundance (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *blaSHV* | − | + | + | + | + | + | + | + | + | + | 9.0 |
| *blaTEM* | + | + | + | + | + | + | + | + | + | + | 10 |
| *blaCTX* | − | + | + | − | − | − | − | − | − | + | 3.0 |
| *tet-X* | + | + | + | + | + | + | − | − | + | − | 7.0 |
| *van-A* | + | + | + | + | + | + | + | + | + | + | 10 |
| *mcr1* | + | + | + | + | + | + | + | + | + | + | 10 |
| *aac3-II* | − | − | − | − | + | + | + | − | − | + | 4.0 |
| *intI1* | − | + | + | + | + | + | + | + | − | − | 7.0 |
| *intI2* | + | − | + | + | + | + | − | − | + | − | 6.0 |

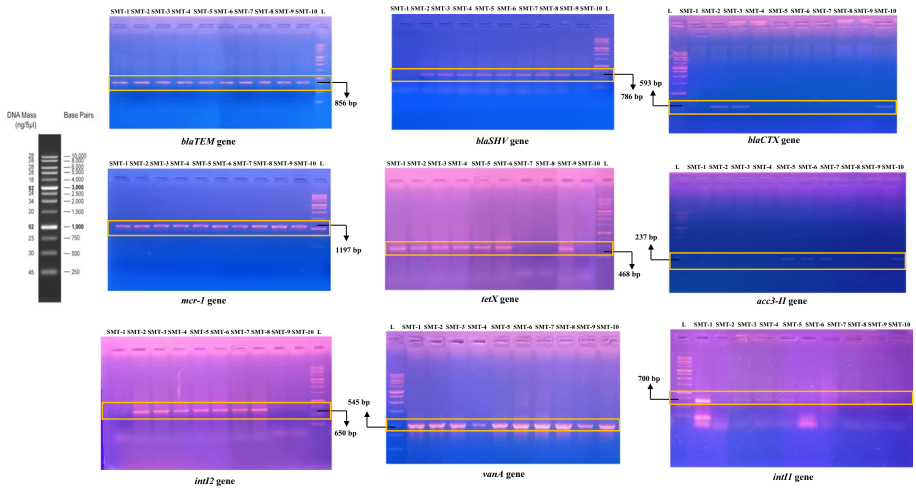

**Fig. 5** Amplification of signature sequences of different ARGs (*blaTEM*, *blaSHV*, *blaCTX*, *mcr-1*, *tetX*, *aac3-II*, *int2*, *vanA*, and *intI1*) from the metagenomes of respective SMT products

[28]. Therefore, it encouraged us to assess the antibiotic resistance profile of inhabitant bacteria of SMT products. The major aim of this investigation was to explore the bacterial diversity of indigenous smokeless tobacco samples and understand the correlation with antibiotic-resistant genes or pathways.

Bacterial Diversity. Phylum-level analysis revealed that commercial smokeless tobacco products are dominated by Firmicutes and Proteobacteria followed by Actinobacteria. This profile follows our previous investigation on smokeless tobacco products and corroborates the other reports on smokeless tobacco bacterial microbiome [26, 29]. However, Sajid et al. [30] obtained a higher count of Proteobacteria in Indian smokeless tobacco products (Table 6). It may be due to the moist smokeless tobacco samples which support bacterial growth. Genus-level analysis showed the prevalence of *Bacillus* spp. among the tobacco samples studied here. It is consistent with the previous reports on smokeless tobacco products [26, 31, 32], where *Bacillus* was observed as the dominant genus. We observed only one genus, i.e., *Dickeya* (*Erwinia*), which was the only Proteobacteria among the top ten genera. *Dickeya* is a well-known Gram-negative plant pathogen and finds a role in biofilm formation due to the production of putrescine, a polyamine. Though *Dickeya* has never been reported for causing diseases among humans, regulation of this genus in tobacco products will be better to improve the quality of commercial smokeless tobacco products. The occurrence of *Corynebacterium* and *Staphylococcus* in smokeless tobacco products may be correlated with their elevated count in oral cancer patients [33]. Srivastava et al. observed a higher count of *Corynebacterium* in the oral cavity of tobacco chewers [9]. These bacteria find a role in nitrate/nitrite reduction and must be confirmed for their role in tobacco-specific nitrosamine (TSNA) formation [34–36]. TSNAs have been reported for their key role in cancer initiation [10]. *Pseudomonas* was also observed among all the tobacco samples studied here which supports the colonization activity due to several factors such as biofilm formation [37], quorum sensing activity [38], and mucoid development [39]. Species-level analysis follows the pattern of genus-level analysis, with the prevalence of *Bacillus* spp., and it follows the pattern of previous research on smokeless tobacco products [26, 29, 30]. In our previous investigation, we observed a significantly higher count of

**Table 6** Previous investigations on metagenome-based analysis on SMT products

| Country | Sample types | NGS platform | OTU/ASV picking platform | Top five phyla | Top five genera | Top five species | Functional annotation | Antibiotic resistance pathways | References |
|---|---|---|---|---|---|---|---|---|---|
| USA | Dry and moist snuff, Swedish-made snus, and Sudanese toombak | Ion torrent PGM | QIIME version 1.8.0 | Actinobacteria, Firmicutes, Proteobacteria, Bacteroidetes | *Bacillus*, *Corynebacterium*, *Staphylococcus*, *Pseudomonas*, *Tetragenococcus* | -NM- | KEGG Orthology | Bicyclomycin/chloramphenicol resistance | [31] |
| Multinational products | American moist snuff, Swedish, Sudanese toombak, Yemeni shammah | Illumina MiSeq sequencing | Mothur version 1.38.1 | Firmicutes, Proteobacteria, Actinobacteria, Cyanobacteria, Bacteroidetes | *Bacillus*, *Paenibacillus*, *Massilia*, *Pseudomonas*, *Candidatus* | *B. safensis*, *B. pumilus*, *Stratosphericus altitudinis*, *B. clausii* | KEGG orthologs | -NM- | [26] |
| USA | -NM- | Illumina, Inc., San Diego, CA | SPADES Meta v3.10.0 | -NM- | -NM- | -NM- | CARD, ICEBERG | -NM- | [41] |
| North India | Chewable tobacco, snus, and snuff | MinION Platform | Albacore Software v2.0.1 | Firmicutes, Actinobacteria, Proteobacteria, Bacteroidetes, Fusobacteria | *Staphylococcus*, *Bacillus*, *Corynebacterium*, *Prevotella*, *Pantoea* | *S. pasteuri*, *S. epidermidis*, *B. pumilus*, *Virgibacillus halodentrificans*, *Virgibacillus marismortoui* | -NM- | -NM- | [29] |

**Table 6** (continued)

| Country | Sample types | NGS platform | OTU/ASV picking platform | Top five phyla | Top five genera | Top five species | Functional annotation | Antibiotic resistance pathways | References |
|---|---|---|---|---|---|---|---|---|---|
| Indian region | Khaini, moist snuff, Qiwam, Snus | Illumina MiSeq sequencing | QIIME version 1.9.0 | Proteobacteria, Firmicutes, Actinobacteria, Bacteroidetes | *Acinetobacter*, *Prevotella*, *Faecalibacterium*, *Bacillus*, *Lactobacillus* | -NM- | KEGG orthology | Multiple antibiotic resistance protein, penicillin-binding protein | [11] |
| USA | Commercial brands and toombak | 454 FLX Titanium pyrosequencing | QIIME release v. 1.9 | Firmicutes, Proteobacteria, Actinobacteria, Bacteroidetes | *Acetobacter*, *Kurthia*, *Lactobacillus*, *Ralstonia*, *Burkholderia* | -NM- | KEGG orthology | Clavulanic acid, penicillin, cephalosporin, novobiocin, streptomycin, tetracycline, and vancomycin biosynthesis pathways | [42] |
| Indian region | Commercial and loose leaf | Illumina MiSeq | QIIME version 1.9.0 | Actinobacteria, Bacteroidetes, Firmicutes, Proteobacteria, Acidobacteria | *Acinetobacter*, *Bacteroides*, *Prevotella*, *Lactobacillus*, *Ruminococcus* | -NM- | KEGG orthology | Penicillin-binding protein, multiple antibiotic resistance protein, macrolide phosphotransferase, beta-lactamase classes | [43] |

**Table 6** (continued)

| Country | Sample types | NGS platform | OTU/ASV picking platform | Top five phyla | Top five genera | Top five species | Functional annotation | Antibiotic resistance pathways | References |
|---|---|---|---|---|---|---|---|---|---|
| Sudan | Toombak | Illumina MiSeq device 2×300 | QIIME | Firmicutes, Actinobacteria, Cyanobacteria | *Corynebacterium_1*, *Atopostipes*, *Atopococcus*, *Oceanobacillus*, *Yaniella* | *Corynebacterium casei*, *Atopococcus tabaci*, *Atopostipes suicloacalis*, *Oceanobacillus chironomi*, and *Staphylococcus gallinarum* | KEGG orthology | ATP binding protein, quinone reductase | [44] |
| North Indian region | Chewable tobacco, snus, and moist snuff | Illumina HiSeq 2500 | QIIME2 pipeline (ver.2020.11) | Firmicutes, Proteobacteria, Actinobacteria, Bacteroidetes, Cyanobacteria | *Lysinibacillus*, *Bacillus*, *Terribacillus*, *Dickeya*, *Oceanobacillus* | *Dickeya phage*, *Lysinibacillus xylanilyticus*, *Bacterium B16S*, *Oceanobacillus caeni*, *Staphylococcus carnosus* | KEGG and BRITE database | CAMP, vancomycin, penicillin-binding protein, multidrug resistance MDR, β-lactam, tetracycline | Present investigation |

*NM* not mentioned

*Bacillus* spp. among commercial tobacco products [10]. Of the various samples, SMT-1 and SMT-4 showed greater values of diversity indices over the other samples. It may be due to the higher moisture content which allows more bacteria to survive in tobacco leaves. However, β-diversity among the samples showed clustering among most of the samples during weighted_unifrac which indicates a consistent diversity as we observed in our previous investigation [10]. *Bacillus* spp. contributes to nitrate reduction and may assist TSNA formation through nitrosation [10, 31, 40].

**Antibiotic Resistance Genes** The increasing threat of antibiotic-resistant bacteria is a big concern especially in food and beverage human consumption [45]. Super resistance capability of bacteriomes against various classes of antibiotics increases the rate of mortality and increases the economic cost [46]. Here, we attempted to correlate the bacterial diversity of commercial smokeless tobacco products with the predicted ARG profiles. Handful reports discuss the ARG profile of inhabitant bacteria of smokeless tobacco products [41, 42, 44]. Only two reports are available from India in a similar line that discusses ARG pathways, and thus, this investigation further adds information on the ARG profile from Indian SMT samples (Table 6). The occurrence of several common antibiotic resistance mechanisms such as vancomycin, penicillin-binding protein, multidrug resistance MDR, and β-lactam corroborates the present findings with previous investigations on smokeless tobacco products [11, 43]. Nevertheless, we also observed the significant dominance of CAMP pathways (~33%) among the samples which is the first time observed in the Indian tobacco. Other reports from Indian SMT samples did not report the prevalence of CAMP pathways [30, 43]. Therefore, it encourages understanding the adapted mechanisms for CAMP among the inhabiting bacteria of SMT samples. Cross-linking of cell envelope components, release of CAMP trapping molecules, efflux of CAMPs, and secretion of peptidases are the few mechanisms reported to degrade CAMP antibiotics [47–50]. *Paenibacillus*, *Terribacillus*, and *Halomonas* showed a significantly positive correlation with the CAMP-associated ARGs. These genera were prevalent among Indian SMT samples and thus need to be further investigated for CAMP resistance using a culturable approach. Besides, several ARGs were predicted among dominant genera. *Paenibacillus* and other *Bacillus* spp. have previously been reported for antibiotic resistance in food products such as honey [51]. Different genera of Firmicutes have been reported for carrying high loads of ARGs [52]. *Aerococcus* also showed antibiotic resistance for various ARGs here, which are generally sensitive to β-lactam antibiotics along with cefotaxime and ceftriaxone [53, 54]. *Aerococcus*, an opportunistic bacterium, has been reported as one of the prevalent genera of smokeless tobacco [6, 31, 42]. *Alkalibacterium* is a member of lactic acid bacteria (LAB) which assist in fermentation and is considered safe bacteria for the human gut [55]. However, here, it showed resistance to several antibiotics. The prevalence of ARG resistance *Alkalibacterium* in commercial smokeless tobacco samples may encourage the dissemination of antibiotic-resistant genes. *Dickeya* and *Halomonas* belong to Gram-negative bacteria which showed antibiotic resistance to many antibiotics. The bacteria *Staphylococcus*, *Halomonas*, and *Corynebacterium* were observed to be resistant to bicyclomycin/chloramphenicol and vancomycin like antibiotics [31]. A recent study on whole metagenome sequencing of smokeless tobacco products unveiled the presence of several ARGs such as β-lactam, penicillin, vancomycin, macrolides, and aminoglycoside antibiotics, as well as genes encoding for multidrug transporters and efflux pumps in SMT samples [56]. Whole metagenome sequencing should be more explored to enrich the limited information of ARGs on SMT products. The spread of these genes among smokeless tobacco chewers poses a significant health threat. We also observed *Pseudomonas* and *Lysinibacillus*; however, these genera were negatively and significantly correlated with the prevalent ARGs predicted here. It may be due to their less abundance (< 0.01). However, more investigation is required on *Pseudomonas* spp. as this genus exhibits strong biofilm formation activity along with resistance against several antibiotics such as quinolones and β-lactams [57].

An attempt was also made to amplify some signature sequences of ARGs (*blaTEM*, *blaSHV*, *blaCTX*, *tetX*, *vanA*, *aac3-II*, *mcr-1*, *intI-1*, and *intI2*) which showed their abundance in worldwide SMT samples [58]. All ARGs here encode a type of enzymatic-modified antibiotic, which can be easily disseminated [59, 60]. These genes exhibited either intrinsic or extrinsic behavior through spontaneous mutation. It could be possible due to horizontal gene transfer via conjugative plasmids, transposons, and the possession of integron and insertion elements [46, 59]. Therefore, smokeless products exaggerate the health risks by disseminating the ARGs among the commensal bacteria of the human oral cavity as well as the gut.

## Conclusion

The study concludes that Indian commercial smokeless tobacco is rich in Firmicutes having a prevalence of genus *Bacillus*. ARG profiling of smokeless tobacco bacteriome showed the supremacy of CAMP resistance genes followed by vancomycin, penicillin-binding protein, and multidrug resistance among the top 10% ARGs. Correlation network analysis unveiled a significantly strong positive correlation between abundant genera and several ARGs. *Staphylococcus*, *Paenibacillus*, and *Aerococcus* were observed for the presence of several ARGs. It includes genes of CAMP, β-lactam, vancomycin, tetracycline, and aminoglycosides. The introduction of antibiotic-resistant bacteria along with smokeless tobacco products during consumption may enhance the possibility of disseminating antibiotic resistance genes in the oral cavity. It may alter the oral microbiome and enhance the oral health risks. Therefore, extensive research is required in line with culture-dependent approaches to investigate the ARG threat on commercial SMT products. Shotgun approaches can be better employed to understand the dynamics between microorganisms and antibiotic resistance genes.

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**Author Contribution** DV conceived the idea and designed the analysis of the present work. AV performed the NGS-based analysis and generated the analytical graphs. AV and DV wrote the manuscript. Both the authors read and approved the final version of the manuscript.

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**Data Availability** Illumina HiSeq data of this investigation has been submitted to the NCBI Sequence Read Archive (SRA) database under the BioProject accession number PRJNA1033989.

## Declarations

**Ethical Approval** Not applicable.

**Consent to Participate** Not applicable.

**Consent for Publication** On acceptance of the manuscript the copyright from the authors will be transferred to the journal.

**Competing Interests** The authors declare no competing interests.

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